Yifei Dong

Ph.D. Student, University of Washington
Tel: +1-917-355-6197
Email: yfeidong@uw.edu
Homepage: https://f1y1113.github.io/

Short Biography

Yifei Dong is a Ph.D. student at the University of Washington, advised by Prof. Zhi-Qi Cheng. He previously worked as a research assistant at Carnegie Mellon University and a data scientist intern at Graphen. His research focuses on embodied AI, with an emphasis on world models and vision-and-language navigation, integrating language grounding, visual imagination, and planning. He received his M.S. in Electrical Engineering from Columbia University. His first/co-first-author publications include NeurIPS and EMNLP Oral presentations, as well as papers at ECCV, IROS, and ACL.

Education

University of Washington USA
Ph.D. in Computer Science and Systems; Advisor: Prof. Zhi-Qi Cheng *Sep 2025 - Present*

Columbia University USA
M.S. in Electrical Engineering *Feb 2023*
Specialization: Data-Driven Analysis and Computation.
Coursework: Machine Learning, Advanced Computer Vision, Database, Statistical Learning, Algorithms.

University of Glasgow UK
B.S. in Electronic & Electrical Engineering *Jun 2021*
Honours of the Second Class (Division i); Outstanding Graduation Design.

Awards

- **Carwein-Andrews PhD Fellowship**, UW, 2025-2026 ($15,000 annual stipend; 2 recipients annually).
- **Top Scholar Award**, UW Graduate School Fund for Excellence and Innovation (GSFEI), 2025.
- **Oral Presentation**, NeurIPS 2026 (first author; top 0.36% of submissions).
- **Top Reviewer**, NeurIPS 2026.
- **Outstanding Paper Finalist and Oral Presentation**, EMNLP 2024 (top 4.8% of submissions).
- **Best Paper Award** (1 of 2), CVPR 2025 Anti-UAV Workshop.
- **Financial Assistance Award**, NeurIPS 2026.
- **DEI Travel Grant**, ECCV 2026.
- **IEEE RAS Travel Support**, IROS 2026.
- **Graduate School Conference Presentation Award**, UW, 2026.
- **1st Place**, NIST TRECVID 2022 Deep Video Understanding Challenge (Oral).

Service

- *Competition Organizer:* RoboWorld Challenge, associated with RoboPAD @ NeurIPS 2026.
- *Reviewer:* TPAMI; AAAI (2026, 2027), CVPR (2026), ECCV (2026), IROS (2026), NeurIPS (2026), ICRA (2027).
- *Teaching Assistant:* UW TCSS 458: Computer Graphics (Spring 2026)

Experience

Research Assistant: Multimodal Intelligence Lab, University of Washington Sep 2025 - Present
Advisor: Prof. Zhi-Qi Cheng; Mentors: Dr. Qi Dai and Prof. Alex Hauptmann.

- Led the development and open-source release of **LCVN**, exploring two paradigms for language-conditioned navigation: policy learning in imagined latent space and unified autoregressive action–observation prediction. The work was selected for a **NeurIPS 2026 Oral** (112/30,709 submissions); LCVN serves as the benchmark for the **WorldNav track** of RoboWorld Challenge 2026.
- Introduced **UniWM**, the first unified, memory-augmented world model to integrate visual navigation planning and imagination within a single multimodal autoregressive backbone. UniWM and LCVN have collectively attracted **530+** GitHub stars.

Research Assistant: Carnegie Mellon University Oct 2022 - Jul 2025
Mentors: Prof. Zhi-Qi Cheng, Dr. Qi Dai, Prof. Kate Whitefoot, and Prof. Alex Hauptmann.

- Led **HA-VLN** (**520+** GitHub stars), releasing simulation environments and evaluation tools for robot navigation around moving people; HA-VLN 2.0 serves as the benchmark for the **human-aware navigation track** of RoboWorld Challenge 2026.

- Developed SHIELD for the Manufacturing Futures Institute (**MFI**) project, Digital Twins for Manufacturing Resiliency to Material Supply Disruptions, combining LLMs and domain expertise to identify and assess EV battery supply chain disruption risks.

Data Scientist Intern: Graphen, Inc. Feb 2022 - Dec 2022
Mentor: Dr. Ching-Yung Lin.

- Developed language-image pre-training methods for long-form video understanding, making core contributions to Graphen's **1st-place finish worldwide** in the **NIST TRECVID** 2022 Deep Video Understanding Evaluation.

SELECTED PUBLICATIONS

10 selected publications (**7 as first/co-first author**) and 1 preprint. * equal contribution

World Models & Embodied Navigation

- **Language-Conditioned World Modeling for Visual Navigation**
  Yifei Dong*, Fengyi Wu*, Yilong Dai, Lingdong Kong, <3 more authors>, Feng Liu, Siyu Huang, Qi Dai, and Zhi-Qi Cheng
  NeurIPS 2026 Oral; 0.36% of submissions (112/30,709)

- **Towards Unified World Models for Visual Navigation via Memory-Augmented Planning and Foresight**
  Yifei Dong*, Fengyi Wu*, Guangyu Chen, Lingdong Kong, <5 more authors>, Qi Dai, Alexander G. Hauptmann, and Zhi-Qi Cheng
  ECCV 2026

- **HA-VLN 2.0: An Open Benchmark and Leaderboard for Human-Aware Navigation in Discrete and Continuous Environments with Dynamic Multi-Human Interactions**
  Yifei Dong*, Fengyi Wu*, Qi He, Lingdong Kong, <5 more authors>, Qi Dai, Alexander G. Hauptmann, and Zhi-Qi Cheng
  IROS 2026

- **GoViG: Goal-Conditioned Visual Navigation Instruction Generation via Multimodal Reasoning**
  Fengyi Wu*, Yifei Dong*, <5 more authors>, Qi Dai, Alexander G. Hauptmann, and Zhi-Qi Cheng
  ACL 2026 Findings

- **Human-Aware Vision-and-Language Navigation: Bridging Simulation to Reality with Dynamic Human Interactions**
  Heng Li*, Minghan Li*, Zhi-Qi Cheng*, Yifei Dong, Yuxuan Zhou, Jun-Yan He, Qi Dai, Teruko Mitamura, and Alexander G. Hauptmann
  NeurIPS 2024 Spotlight

- **Tiresias: Towards Language-Conditioned World Modeling**
  Fengyi Wu*, Yifei Dong*, Yetong Sha, <3 more authors>, Qi Dai, Alexander G. Hauptmann, and Zhi-Qi Cheng
  Preprint, under review

Vision, Language & LLM Applications

- **SHIELD: LLM-Driven Schema Induction for Predictive Analytics in EV Battery Supply Chain Disruptions**
  Zhi-Qi Cheng*, Yifei Dong*, <3 more authors>, Jason O'Connor, Alexander G. Hauptmann, and Kate Whitefoot
  EMNLP 2024 Oral; Outstanding Paper Finalist

- **Large Language Model Agents in Finance: A Survey Bridging Research, Practice, and Real-World Deployment**
  Yifei Dong*, Fengyi Wu*, Kunlin Zhang, Yilong Dai, Sanjian Zhang, Wanghao Ye, Sihan Chen, and Zhi-Qi Cheng
  EMNLP 2025 Findings

- **Securing the Skies: A Comprehensive Survey on Anti-UAV Methods, Benchmarking, and Future Directions**
  Yifei Dong*, Fengyi Wu*, <4 more authors>, Jingdong Sun, Siyu Huang, Feng Liu, Qi Dai, and Zhi-Qi Cheng
  CVPR Anti-UAV Workshop 2025 Best Paper

- **MaxSup: Overcoming Representation Collapse in Label Smoothing**
  Yuxuan Zhou*, Heng Li*, Zhi-Qi Cheng, Xudong Yan, Yifei Dong, Mario Fritz, and Margret Keuper
  NeurIPS 2025 Oral

- **Semantic Understanding and Evolving Interaction Tracking in Long-form Multimodal Datasets**
  Vishal Anand, Yifei Dong, Raksha Ramesh, Zifan Chen, Yun Chen, Linquan Li, and Ching-Yung Lin
  TRECVID 2022 Oral